1) a)

$f(x) = \sin x$, $f'(x) = \cos x$, $f''(x) = -\sin x$, $f^{(3)}(x) = -\cos x$, $f^{(4)}(x) = \sin x$

1) b)

$f(x) = \cos x$, $f'(x) = -\sin x$, $f''(x) = -\cos x$, $f^{(3)}(x) = \sin x$, $f^{(4)}(x) = \cos x$

1) c)

$f(x) = e^x$, $f'(x) = e^x$, $f''(x) = e^x$, $f^{(3)}(x) = e^x$, $f^{(4)}(x) = e^x$

1) d)

$f(x) = \ln(1+x)$, $f'(x) = (1+x)^{-1}$, $f''(x) = -(1+x)^{-2}$, $f^{(3)}(x) = 2(1+x)^{-3}$,
$f^{(4)}(x) = -6(1+x)^{-4}$

1) e)

$f(x) = (1+x)^n$, $f'(x) = n(1+x)^{n-1}$, $f''(x) = n(n-1)(1+x)^{n-2}$,
$f^{(3)}(x) = n(n-1)(n-2)(1+x)^{n-3}$, $f^{(4)}(x) = n(n-1)(n-2)(n-3)(1+x)^{n-4}$

2) a)

$f(x) = \sin x$, $f'(x) = \cos x$, $f''(x) = -\sin x$, $f^{(3)}(x) = -\cos x$, $f^{(4)}(x) = \sin x$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = -1$, $f^{(4)}(0) = 0$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}}{r!}x^r$$

So $\sin x = 0 + x + 0 - \frac{1}{6}x^3 + 0 + \cdots$

$\therefore \sin x \approx x - \frac{1}{6}x^3$

2) b)

$f(x) = \cos x$, $f'(x) = -\sin x$, $f''(x) = -\cos x$, $f^{(3)}(x) = \sin x$, $f^{(4)}(x) = \cos x$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 1$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}}{r!}x^r$$

So $\cos x = 1 + 0 - \frac{1}{2}x^2 + 0 + \frac{1}{24}x^4 + \cdots$

$\therefore \cos x \approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

2) c)

$f(x) = e^x$, $f'(x) = e^x$, $f''(x) = e^x$, $f^{(3)}(x) = e^x$, $f^{(4)}(x) = e^x$

So $f(0) = 1$, $f'(0) = 1$, $f''(0) = 1$, $f^{(3)}(0) = 1$, $f^{(4)}(0) = 1$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots$

$\therefore e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4$

2) d)

$f(x) = \ln(1 + x)$, $f'(x) = (1 + x)^{-1}$, $f''(x) = -(1 + x)^{-2}$, $f^{(3)}(x) = 2(1 + x)^{-3}$, $f^{(4)}(x) = -6(1 + x)^{-4}$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f^{(3)}(0) = 2$, $f^{(4)}(0) = -6$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln(1 + x) = 0 + x - \frac{1}{2}x^2 + \frac{2}{6}x^3 - \frac{6}{24}x^4 + \cdots$

$\therefore \ln(1 + x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4$

2) e)

$f(x) = (1 + x)^n$, $f'(x) = n(1 + x)^{n-1}$, $f''(x) = n(n - 1)(1 + x)^{n-2}$, $f^{(3)}(x) = n(n - 1)(n - 2)(1 + x)^{n-3}$, $f^{(4)}(x) = n(n - 1)(n - 2)(n - 3)(1 + x)^{n-4}$

So $f(0) = 1$, $f'(0) = n$, $f''(0) = n(n - 1)$, $f^{(3)}(0) = n(n - 1)(n - 2)$, $f^{(4)}(0) = n(n - 1)(n - 2)(n - 3)$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $(1 + x)^n = 1 + x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \cdots$

$\therefore (1 + x)^n \approx 1 + x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4$

3)

Let $f(x) = \ln x$, then $f(0)$ is undefined.

Additionally, $f'(x) = \frac{1}{x}$ and so $f'(0)$ does not exist for $f(x) = \ln x$ (and the higher derivatives also do not exist at $x = 0$)

$\therefore$ the Maclaurin series cannot be found.

4) a)

$\mathrm{f}(x) = \mathrm{e}^{5x}$, $\mathrm{f}'(x) = 5\mathrm{e}^{5x}$, $\mathrm{f}''(x) = 25\mathrm{e}^{5x}$

So $\mathrm{f}(0) = 1$, $\mathrm{f}'(0) = 5$, $\mathrm{f}''(0) = 25$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\mathrm{e}^{5x} = 1 + 5x + \frac{25}{2}x^2 + \cdots$

$\therefore \mathrm{e}^{5x} \approx 1 + 5x + \frac{25}{2}x^2$ valid for all x

4) b)

$\mathrm{f}(x) = \sin 3x$, $\mathrm{f}'(x) = 3\cos 3x$, $\mathrm{f}''(x) = -9\sin 3x$, $\mathrm{f}^{(3)}(x) = -27\cos 3x$,

$\mathrm{f}^{(4)}(x) = 81\sin 3x$, $\mathrm{f}^{(5)}(x) = 243\cos 3x$

So $\mathrm{f}(0) = 0$, $\mathrm{f}'(0) = 3$, $\mathrm{f}''(0) = 0$, $\mathrm{f}^{(3)}(0) = -27$, $\mathrm{f}^{(4)}(0) = 0$, $\mathrm{f}^{(5)}(0) = 243$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\sin 3x = 0 + 3x - 0 - \frac{27}{6}x^3 + 0 + \frac{243}{120}x^5 + \cdots$

$\therefore \sin 3x \approx 3x - \frac{9}{2}x^3 + 0 + \frac{81}{40}x^5$ valid for all x

4) c)

$\mathrm{f}(x) = \cos\frac{x}{2}$, $\mathrm{f}'(x) = -\frac{1}{2}\sin\frac{x}{2}$, $\mathrm{f}''(x) = -\frac{1}{4}\cos\frac{x}{2}$, $\mathrm{f}^{(3)}(x) = \frac{1}{8}\sin\frac{x}{2}$, $\mathrm{f}^{(4)}(x) = \frac{1}{16}\cos\frac{x}{2}$

So $\mathrm{f}(0) = 1$, $\mathrm{f}'(0) = 0$, $\mathrm{f}''(0) = -\frac{1}{4}$, $\mathrm{f}^{(3)}(0) = 0$, $\mathrm{f}^{(4)}(0) = \frac{1}{16}$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\cos\frac{x}{2} = 1 + 0 + \frac{-\frac{1}{4}}{2}x^2 + 0 + \frac{\frac{1}{16}}{24}x^4 + \cdots$

$\therefore \cos\frac{x}{2} \approx 1 - \frac{1}{8}x^2 + \frac{1}{384}x^4$ valid for all x

4) d)

$f(x) = \ln(1+4x)$, $f'(x) = 4(1+4x)^{-1}$, $f''(x) = -16(1+4x)^{-2}$,

$f^{(3)}(x) = 128(1+4x)^{-3}$

So $f(0) = 0$, $f'(0) = 4$, $f''(0) = -16$, $f^{(3)}(x) = 128$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln(1+4x) = 0 + 4x - \frac{16}{2}x^2 + \frac{128}{6}x^3 + \cdots$

$\therefore \ln(1+4x) \approx 4x - 8x^2 + \frac{64}{3}x^3$

Validity: $-1 < 4x \leq 1 \Rightarrow -\frac{1}{4} < x \leq \frac{1}{4}$

$\therefore \ln(1+4x) \approx 4x - 8x^2 + \frac{64}{3}x^3$ valid for $-\frac{1}{4} < x \leq \frac{1}{4}$

4) e)

$f(x) = \ln(x+2)$, $f'(x) = (x+2)^{-1}$, $f''(x) = -(x+2)^{-2}$

So $f(0) = \ln 2$, $f'(0) = \frac{1}{2}$, $f''(0) = -\frac{1}{4}$,

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln(x+2) = \ln 2 + \frac{1}{2}x + \frac{-\frac{1}{4}}{2!}x^2 + \cdots$

$\therefore \ln(x+2) \approx \ln 2 + \frac{1}{2}x - \frac{1}{8}x^2$

$x + 2 = 2\left(1 + \frac{x}{2}\right)$ so validity: $-1 < \frac{x}{2} \leq 1 \Rightarrow -2 < x \leq 2$

$\therefore \ln(x+2) \approx \ln 2 + \frac{1}{2}x - \frac{1}{8}x^2$ valid for $-2 < x \leq 2$

4) f)

$f(x) = (1+2x)^{\frac{1}{4}}$, $f'(x) = \frac{1}{2}(1+2x)^{-\frac{3}{4}}$, $f''(x) = -\frac{3}{4}(1+2x)^{-\frac{7}{4}}$

So $f(0) = 1$, $f'(0) = \frac{1}{2}$, $f''(0) = -\frac{3}{4}$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $(1+2x)^{\frac{1}{4}} = 1 + \frac{1}{2}x - \frac{3}{8}x^2 + \cdots$

$\therefore (1+2x)^{\frac{1}{4}} \approx 1 + \frac{1}{2}x - \frac{3}{8}x^2$ valid for $|x| < \frac{1}{2}$

4) g)

$f(x) = (1-2x)^{-3}$, $f'(x) = 6(1-2x)^{-4}$, $f''(x) = 48(1-2x)^{-5}$

So $f(0) = 1$, $f'(0) = 6$, $f''(0) = 48$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $(1-2x)^{-3} = 1 + 6x + \frac{48}{2}x^2 + \cdots$

Validity: $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$

$\therefore \frac{1}{(1-2x)^3} \approx 1 + 6x + 24x^2$ valid for $|x| < \frac{1}{2}$

5) a)

$f(x) = \cosh x$, $f'(x) = \sinh x$, $f''(x) = \cosh x$, $f^{(3)}(x) = \sinh x$, $f^{(4)}(x) = \cosh x$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = 1$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 1$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\cosh x = 1 + 0 + \frac{1}{2}x^2 + 0 + \frac{1}{24}x^4 + \cdots$

$\therefore \cosh x \approx 1 + \frac{1}{2}x^2 + \frac{1}{24}x^4$

5) b)

$f(x) = \sinh x$, $f'(x) = \cosh x$, $f''(x) = \sinh x$, $f^{(3)}(x) = \cosh x$, $f^{(4)}(x) = \sinh x$, $f^{(5)}(x) = \cosh x$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = 1$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 1$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\sinh x = 0 + x + 0 + \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots$

$\therefore \sinh x \approx x + \frac{1}{6}x^3 + \frac{1}{120}x^5$

6) a)

$f(x) = (1+4x)^{\frac{1}{2}}$

$f'(x) = 2(1+4x)^{-\frac{1}{2}}$ and $f''(x) = -4(1+4x)^{-\frac{3}{2}}$

6) b)

$f(0) = 1, f'(0) = 2, f''(0) = -4$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $(1+4x)^{\frac{1}{2}} = 1 + 2x - 2x^2 + \cdots$

$\therefore \sqrt{1+4x} \approx 1 + 2x - 2x^2$

6) c)

valid for $|x| < \frac{1}{4}$

6) d)

$1 + 4x = 6 \Rightarrow x = \frac{5}{4}$ which is not valid for this expansion

$\sqrt{6} = \sqrt{4\left(1+\frac{1}{2}\right)} = 2\sqrt{1+\frac{1}{2}}$ and so use $x = \frac{1}{8}$

$\sqrt{1+\frac{1}{2}} \approx 1 + 2\left(\frac{1}{8}\right) - 2\left(\frac{1}{8}\right)^2 = \frac{39}{32}$

And so $\sqrt{6} = 2\sqrt{\frac{3}{2}} \approx 2 \times \frac{39}{32} = \frac{39}{16}$ as req.

7) a)

$f(x) = \ln\left(\frac{x+2}{x+1}\right) = \ln(x+2) - \ln(x+1)$ so $f(0) = \ln 2$

$f'(x) = \frac{1}{x+2} - \frac{1}{x+1}$ so $f'(0) = -\frac{1}{2}$

$f''(x) = -\frac{1}{(x+2)^2} + \frac{1}{(x+1)^2}$ so $f''(0) = \frac{3}{4}$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\ln\left(\frac{x+2}{x+1}\right) = \ln 2 - \frac{1}{2}x + \frac{\frac{3}{4}}{2!}x^2 + \cdots$

$\therefore \ln\left(\frac{x+2}{x+1}\right) \approx \ln 2 - \frac{1}{2}x + \frac{3}{8}x^2$

7) b)

Consider $\ln(x+2) = \ln\left(2\left(1+\frac{x}{2}\right)\right) = \ln 2 + \ln\left(1+\frac{x}{2}\right)$

Validity: $-1 < \frac{x}{2} \leq 1 \Rightarrow -2 < x \leq 2$

Consider $\ln(x+1)$ which has validity $-1 < x \leq 1$

$\therefore -1 < x \leq 1$

8) a)

$\sin\left(x-\frac{\pi}{4}\right) = \sin x \cos\frac{\pi}{4} - \cos x \sin\frac{\pi}{4}$

$= \frac{\sqrt{2}}{2}\left(x - \frac{x^3}{3!} + \cdots\right) - \frac{\sqrt{2}}{2}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)$

$= \frac{\sqrt{2}}{2}\left(-1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right)$ as req.

8) b)

$\mathrm{f}(x) = \sin\left(x-\frac{\pi}{4}\right)$ so $\mathrm{f}(0) = -\frac{\sqrt{2}}{2}$

$\mathrm{f}'(x) = \cos\left(x-\frac{\pi}{4}\right)$ so $\mathrm{f}'(0) = \frac{\sqrt{2}}{2}$

$\mathrm{f}''(x) = -\sin\left(x-\frac{\pi}{4}\right)$ so $\mathrm{f}''(0) = \frac{\sqrt{2}}{2}$

$\mathrm{f}^{(3)}(x) = -\cos\left(x-\frac{\pi}{4}\right)$ so $\mathrm{f}^{(3)}(0) = -\frac{\sqrt{2}}{2}$

$\mathrm{f}^{(4)}(x) = \sin\left(x-\frac{\pi}{4}\right)$ so $\mathrm{f}^{(4)}(0) = -\frac{\sqrt{2}}{2}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sin\left(x-\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}x + \frac{\frac{\sqrt{2}}{2}}{2}x^2 - \frac{\frac{\sqrt{2}}{2}}{6}x^3 - \frac{\frac{\sqrt{2}}{2}}{24}x^4 + \cdots$

$\therefore \sin\left(x-\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}\left(-1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right)$ as req.

9) a)

$\cos\left(x+\frac{3\pi}{4}\right) = \cos x \cos\frac{3\pi}{4} - \sin x \sin\frac{3\pi}{4}$

$= -\frac{\sqrt{2}}{2}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) - \frac{\sqrt{2}}{2}\left(x - \frac{x^3}{3!} + \cdots\right)$

$= \frac{\sqrt{2}}{2}\left(-1 + \frac{x^2}{2} - \frac{x^4}{24} + \cdots\right) - \frac{\sqrt{2}}{2}\left(x - \frac{x^3}{6} + \cdots\right)$

$= \frac{\sqrt{2}}{2}\left(-1 - x + \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right)$ as req.

9) b)

$f(x) = \cos\left(x + \frac{3\pi}{4}\right)$ so $f(0) = -\frac{\sqrt{2}}{2}$

$f'(x) = -\sin\left(x + \frac{3\pi}{4}\right)$ so $f'(0) = -\frac{\sqrt{2}}{2}$

$f''(x) = -\cos\left(x + \frac{3\pi}{4}\right)$ so $f''(0) = \frac{\sqrt{2}}{2}$

$f^{(3)}(x) = \sin\left(x + \frac{3\pi}{4}\right)$ so $f^{(3)}(0) = \frac{\sqrt{2}}{2}$

$f^{(4)}(x) = \cos\left(x + \frac{3\pi}{4}\right)$ so $f^{(4)}(0) = -\frac{\sqrt{2}}{2}$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\cos\left(x + \frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}x + \frac{\frac{\sqrt{2}}{2}}{2!}x^2 + \frac{\frac{\sqrt{2}}{2}}{3!}x^3 - \frac{\frac{\sqrt{2}}{2}}{4!}x^4 + \cdots$

$\therefore \cos\left(x + \frac{3\pi}{4}\right) = \frac{\sqrt{2}}{2}\left(-1 - x + \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right)$ as req.

10) a)

$f(x) = \sin 2x$, $f'(x) = 2\cos 2x$, $f''(x) = -4\sin 2x$, $f^{(3)}(x) = -8\cos 2x$,

So $f(0) = 0$, $f'(0) = 2$, $f''(0) = 0$, $f^{(3)}(0) = -8$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\sin 2x = 0 + 2x + 0 - \frac{8}{3!}x^3 + \cdots$

$\therefore \sin 2x \approx 2x - \frac{4}{3}x^3$

10) b)

$f(x) = \cos 2x$, $f'(x) = -2\sin 2x$, $f''(x) = -4\cos 2x$, $f^{(3)}(x) = 8\sin 2x$,

$f^{(4)}(x) = 16\cos 2x$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -4$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 16$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\cos 2x = 1 + 0 - \frac{4}{2!}x^2 + 0 + \frac{16}{4!}x^4 + \cdots$

$\therefore \cos 2x \approx 1 - 2x^2 + \frac{2}{3}x^4$

10) c)

$\cos\left(2x+\frac{\pi}{6}\right)=\cos 2x\cos\frac{\pi}{6}-\sin 2x\sin\frac{\pi}{6}$

So $\cos\left(2x+\frac{\pi}{6}\right)\approx\left(1-2x^2+\frac{2}{3}x^4\right)\times\frac{\sqrt{3}}{2}-\left(2x-\frac{4}{3}x^3\right)\times\frac{1}{2}$

$=\frac{\sqrt{3}}{2}-x-\sqrt{3}x^2+\frac{2}{3}x^3+\frac{\sqrt{3}}{3}x^4$ as req.

11) a)

$\mathrm{f}(x)=\sin\left(\frac{\pi}{3}+x\right)$ so $\mathrm{f}(0)=\frac{\sqrt{3}}{2}$

$\mathrm{f}'(x)=\cos\left(\frac{\pi}{3}+x\right)$ so $\mathrm{f}'(0)=\frac{1}{2}$

$\mathrm{f}''(x)=-\sin\left(\frac{\pi}{3}+x\right)$ so $\mathrm{f}''(0)=-\frac{\sqrt{3}}{2}$

$\mathrm{f}(x)=\mathrm{f}(0)+\mathrm{f}'(0)x+\frac{\mathrm{f}''(0)}{2!}x^2+\frac{\mathrm{f}^{(3)}(0)}{3!}x^3+\frac{\mathrm{f}^{(4)}(0)}{4!}x^4+\frac{\mathrm{f}^{(5)}(0)}{5!}x^5+\cdots+\frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sin\left(\frac{\pi}{3}+x\right)=\frac{\sqrt{3}}{2}+\frac{1}{2}x+\frac{-\frac{\sqrt{3}}{2}}{2!}x^2+\cdots$

$\therefore\sin\left(\frac{\pi}{3}+x\right)\approx\frac{\sqrt{3}}{2}+\frac{1}{2}x-\frac{\sqrt{3}}{4}x^2$

11) b)

$\int_0^h x^2\sin\left(\frac{\pi}{3}+x\right)\mathrm{d}x\approx\int_0^h x^2\left(\frac{\sqrt{3}}{2}+\frac{1}{2}x-\frac{\sqrt{3}}{4}x^2\right)\mathrm{d}x=\int_0^h\left(\frac{\sqrt{3}}{2}x^2+\frac{1}{2}x^3-\frac{\sqrt{3}}{4}x^4\right)\mathrm{d}x$

$=\left[\frac{\sqrt{3}}{6}x^3+\frac{1}{8}x^4-\frac{\sqrt{3}}{20}x^5\right]_0^h=\left(\frac{\sqrt{3}}{6}h^3+\frac{1}{8}h^4-\frac{\sqrt{3}}{20}h^5\right)-(0)$

$=\frac{\sqrt{3}h^3}{6}+\frac{h^4}{8}-\frac{\sqrt{3}h^5}{20}$ as req.

12)

$\mathrm{f}(x) = \cos 2x$, $\mathrm{f}'(x) = -2\sin 2x$, $\mathrm{f}''(x) = -4\cos 2x$, $\mathrm{f}^{(3)}(x) = 8\sin 2x$,
$\mathrm{f}^{(4)}(x) = 16\cos 2x$, $\mathrm{f}^{(5)}(x) = -32\sin 2x$, $\mathrm{f}^{(6)}(x) = -64\cos 2x$

So $\mathrm{f}(0) = 1$, $\mathrm{f}'(0) = 0$, $\mathrm{f}''(0) = -4$, $\mathrm{f}^{(3)}(0) = 0$, $\mathrm{f}^{(4)}(0) = 16$, $\mathrm{f}^{(5)}(0) = 0$,
$\mathrm{f}^{(6)}(0) = -64$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\cos 2x = 1 + 0 + \frac{-4}{2}x^2 + 0 + \frac{16}{24}x^4 + 0 + \frac{-64}{720}x^6 \ldots$

$\therefore \cos 2x \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6$

Since $\cos(2A) = 2\cos^2 A - 1$ then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$

$\therefore \cos^2 x \approx \frac{1}{2} + \frac{1}{2}\left(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6\right) = 1 - x^2 + \frac{1}{3}x^4 - \frac{2}{45}x^6$

13) a)

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

So $\sin^2\left(\frac{x}{2}\right) = \frac{1}{2} - \frac{1}{2}\cos x$

$\therefore \sin^2\left(\frac{x}{2}\right) \approx \frac{1}{2} - \frac{1}{2}\left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4\right) = \frac{1}{4}x^2 - \frac{1}{48}x^4$

13) b)

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(\frac{1}{x}\sin^2\left(\frac{x}{2}\right)\right) \mathrm{d}x \approx \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(\frac{1}{x}\left(\frac{1}{4}x^2 - \frac{1}{48}x^4\right)\right) \mathrm{d}x = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(\frac{1}{4}x - \frac{1}{48}x^3\right) \mathrm{d}x$$

$$= \left[\frac{1}{8}x^2 - \frac{1}{192}x^4\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = \left(\frac{1}{8}\left(\frac{\pi}{2}\right)^2 - \frac{1}{192}\left(\frac{\pi}{2}\right)^4\right) - \left(\frac{1}{8}\left(\frac{\pi}{3}\right)^2 - \frac{1}{192}\left(\frac{\pi}{3}\right)^4\right) = 0.15 \text{ (2 DP)}$$

14) a)

$\mathrm{f}(x) = \cos 2x$, $\mathrm{f}'(x) = -2\sin 2x$, $\mathrm{f}''(x) = -4\cos 2x$, $\mathrm{f}^{(3)}(x) = 8\sin 2x$,
$\mathrm{f}^{(4)}(x) = 16\cos 2x$, $\mathrm{f}^{(5)}(x) = -32\sin 2x$, $\mathrm{f}^{(6)}(x) = -64\cos 2x$

So $\mathrm{f}(0) = 1$, $\mathrm{f}'(0) = 0$, $\mathrm{f}''(0) = -4$, $\mathrm{f}^{(3)}(0) = 0$, $\mathrm{f}^{(4)}(0) = 16$, $\mathrm{f}^{(5)}(0) = 0$,
$\mathrm{f}^{(6)}(0) = -64$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\cos 2x = 1 + 0 + \frac{-4}{2}x^2 + 0 + \frac{16}{24}x^4 + 0 + \frac{-64}{720}x^6 \ldots$

Giving $\cos 2x \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6$

Since $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ then

$\sin^2 x \approx \frac{1}{2} - \frac{1}{2}\left(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6\right)$

$\therefore \sin^2 x \approx x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6$

14) b)

$\int_0^{\frac{\pi}{2}} \sin^2 x \,\mathrm{d}x = \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)\mathrm{d}x \approx \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\left(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6\right)\right)\mathrm{d}x$

$= \int_0^{\frac{\pi}{2}} \left(x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6\right)\mathrm{d}x = \left[\frac{1}{3}x^3 - \frac{1}{15}x^5 + \frac{2}{315}x^7\right]_0^{\frac{\pi}{2}}$

$= \left(\frac{1}{3}\left(\frac{\pi}{2}\right)^3 - \frac{1}{15}\left(\frac{\pi}{2}\right)^5 + \frac{2}{315}\left(\frac{\pi}{2}\right)^7\right) - (0)$

$= \frac{\pi^3}{24} - \frac{\pi^5}{480} + \frac{\pi^7}{20160}$

$= 0.804$ (3 SF)

14) c)

Since $\cos(2A) = 1 - 2\sin^2 A$ then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$

$\int_0^{\frac{\pi}{2}} \sin^2 x \,\mathrm{d}x = \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)\mathrm{d}x = \left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_0^{\frac{\pi}{2}} = \left(\frac{1}{4}\pi - 0\right) - (0 - 0) = \frac{\pi}{4}$

14) d)

$\frac{\frac{\pi}{4} - \left(\frac{\pi^3}{24} - \frac{\pi^5}{480} + \frac{\pi^7}{20160}\right)}{\frac{\pi}{4}} \times 100 = -2.4\%$ (2 SF)

15) a)

Since $1 - x + \sin x > 0$ then $\sin x > x - 1$

But $\sin x$ satisfies $-1 \leq \sin x \leq 1$ so $\sin x \leq 1 \Rightarrow 1 - x + \sin x \leq 2 - x$

If $x \geq 2$ then $2 - x \leq 0$ so $1 - x + \sin x \leq 0$ so $\mathrm{f}(x)$ is not defined. So $x < 2$

$\therefore k$ can never be greater than 2 as req.

15) b)

$\mathrm{f}(x) = \ln(1 - x + \sin x)$

$\mathrm{f}'(x) = \frac{-1+\cos x}{1-x+\sin x}$

15) c)

$\cos x \approx 1 - \frac{1}{2}x^2$ and $\sin x \approx x$

So that $\frac{-1+\cos x}{1-x+\sin x} \approx \frac{-1+1-\frac{1}{2}x^2}{1-x+x} = -\frac{1}{2}x^2$

$\therefore \mathrm{f}'(x) \approx -\frac{1}{2}x^2 \Rightarrow \mathrm{f}(x) \approx -\frac{1}{6}x^3 + c$

But $\mathrm{f}(0) = \ln(1 - 0 + \sin 0) = 0$ so that $c = 0$

$\therefore \mathrm{f}(x) \approx -\frac{1}{6}x^3$

16) a)

since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$

$\mathrm{f}(x) = \cosh^2 x = \frac{1}{2} + \frac{1}{2}\cosh 2x$

$\mathrm{f}'(x) = \sinh 2x$

$\mathrm{f}''(x) = 2\cosh 2x$

16) b)

Every odd power of the Maclaurin Series includes $\sinh 0$ which is zero.

16) c)

$\mathrm{f}^{(n)} = 2^{n-1}\cosh 0 = 2^{n-1}$

$\frac{\mathrm{f}^{(n)}}{n!} = \frac{2^{n-1}}{n!}$

17) a)

$\mathrm{f}(x) = \sinh x$, $\mathrm{f}'(x) = \cosh x$, $\mathrm{f}''(x) = \sinh x$, $\mathrm{f}^{(3)}(x) = \cosh x$, $\mathrm{f}^{(4)}(x) = \sinh x$, $\mathrm{f}^{(5)}(x) = \cosh x$

So $\mathrm{f}(0) = 0$, $\mathrm{f}'(0) = 1$, $\mathrm{f}''(0) = 0$, $\mathrm{f}^{(3)}(0) = 1$, $\mathrm{f}^{(4)}(0) = 0$, $\mathrm{f}^{(5)}(0) = 1$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\sinh x = 0 + x + 0 + \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots$

$\therefore \sinh x \approx x + \frac{1}{6}x^3 + \frac{1}{120}x^5$

17) b)

$$\sinh^n x = (\sinh x)^n \approx \left(x + \frac{1}{6}x^3 + \frac{1}{120}x^5\right)^n$$

$$= \left(x\left(1 + \frac{1}{6}x^2 + \frac{1}{120}x^4\right)\right)^n = x^n\left(1 + \frac{1}{6}x^2 + \frac{1}{120}x^4\right)^n$$

$$= x^n\left[1 + n\left(\frac{1}{6}x^2 + \frac{1}{120}x^4\right) + \frac{n(n-1)}{2}\left(\frac{1}{6}x^2 + \frac{1}{120}x^4\right)^2 + \cdots\right]$$

$$= x^n\left[1 + n\left(\frac{1}{6}x^2 + \frac{1}{120}x^4\right) + \frac{n(n-1)}{2}\left(\frac{1}{36}x^4 + \cdots\right) + \cdots\right]$$

$$= x^n + \frac{1}{6}nx^{n+2} + \left(\frac{n}{120} + \frac{n(n-1)}{72}\right)x^{n+4} + \cdots$$

$$\approx x^n + \frac{1}{6}nx^{n+2} + \frac{1}{360}(n(5n-2))x^{n+4}$$

18) a)

$\mathrm{f}(x) = \ln(1 + \cos x)$ so $\mathrm{f}(0) = \ln 2$

$\mathrm{f}'(x) = \frac{-\sin x}{1+\cos x}$ so $\mathrm{f}'(0) = 0$

$\mathrm{f}''(x) = \frac{-\cos x(1+\cos x) - \sin^2 x}{(1+\cos x)^2}$ so $\mathrm{f}''(0) = -\frac{1}{2}$

$$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$$

So $\ln(1 + \cos x) = \ln 2 + 0 + \frac{-\frac{1}{2}}{2!}x^2 + \cdots$

$\therefore \ln(1 + \cos x) \approx \ln 2 - \frac{1}{4}x^2$

18) b)

$\ln(1+\cos x)=0 \Rightarrow 1+\cos x=1 \Rightarrow x=\frac{\pi}{2}$ (choosing the smallest possible value of x)

$\ln 2-\frac{1}{4}x^2=0 \Rightarrow x=2\sqrt{\ln 2}$

So that $\frac{\pi}{2}\approx 2\sqrt{\ln 2} \Rightarrow \pi \approx 4\sqrt{\ln 2}$

But since $\ln 2=\ln(1+1)\approx 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots$ then using $\ln 2\approx 1-\frac{1}{2}=\frac{1}{2}$ gives:

$\pi \approx 4\sqrt{\frac{1}{2}}=\frac{4}{\sqrt{2}}=2\sqrt{2}$ as req.

19)

$\mathrm{f}(x)=\sec x$ so $\mathrm{f}(0)=1$

$\mathrm{f}'(x)=\sec x\tan x$ so $\mathrm{f}'(0)=0$

$\mathrm{f}''(x)=\sec^3 x+\sec^2 x\tan^2 x$ so $\mathrm{f}''(0)=1$

$\mathrm{f}(x)=\mathrm{f}(0)+\mathrm{f}'(0)x+\frac{\mathrm{f}''(0)}{2!}x^2+\frac{\mathrm{f}^{(3)}(0)}{3!}x^3+\frac{\mathrm{f}^{(4)}(0)}{4!}x^4+\frac{\mathrm{f}^{(5)}(0)}{5!}x^5+\cdots+\frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sec x=1+0+\frac{1}{2}x^2+\cdots$

$\therefore \sec x\approx 1+\frac{1}{2}x^2$

20) a)

$\mathrm{f}(x)=\cos x$, $\mathrm{f}'(x)=-\sin x$, $\mathrm{f}''(x)=-\cos x$, $\mathrm{f}^{(3)}(x)=\sin x$, $\mathrm{f}^{(4)}(x)=\cos x$

So $\mathrm{f}(0)=1$, $\mathrm{f}'(0)=0$, $\mathrm{f}''(0)=-1$, $\mathrm{f}^{(3)}(0)=0$, $\mathrm{f}^{(4)}(0)=1$

$\mathrm{f}(x)=\mathrm{f}(0)+\mathrm{f}'(0)x+\frac{\mathrm{f}''(0)}{2!}x^2+\frac{\mathrm{f}^{(3)}(0)}{3!}x^3+\frac{\mathrm{f}^{(4)}(0)}{4!}x^4+\cdots+\frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\cos x=1+0-\frac{1}{2}x^2+0+\frac{1}{24}x^4+\cdots$

$\therefore \cos x\approx 1-\frac{1}{2}x^2+\frac{1}{24}x^4$ as req.

20) b)

Since $\cos x \sec x = 1$ then

$\left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots\right)(1 + ax^2 + bx^4 + \cdots) = 1$

So $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + ax^2 - \frac{1}{2}ax^4 + bx^4 + \cdots = 1$

So $-\frac{1}{2} + a = 0 \therefore a = \frac{1}{2}$

So $\frac{1}{24} - \frac{1}{2} \times \frac{1}{2} + b = 0 \therefore b = \frac{5}{24}$

$\therefore a = \frac{1}{2}$ and $b = \frac{5}{24}$

21) a)

$\mathrm{f}(x) = \sin x$, $\mathrm{f}'(x) = \cos x$, $\mathrm{f}''(x) = -\sin x$, $\mathrm{f}^{(3)}(x) = -\cos x$, $\mathrm{f}^{(4)}(x) = \sin x$, $\mathrm{f}^{(5)}(x) = \cos x$

So $\mathrm{f}(0) = 0$, $\mathrm{f}'(0) = 1$, $\mathrm{f}''(0) = 0$, $\mathrm{f}^{(3)}(0) = -1$, $\mathrm{f}^{(4)}(0) = 0$, $\mathrm{f}^{(5)}(0) = 1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sin x = 0 + x + 0 - \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots$

$\therefore \sin x \approx x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

21) b)

Since $\sin x \operatorname{cosec} x = 1$ then

$\left(\frac{1}{x} + ax + bx^3 + \cdots\right)\left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots\right) = 1$

So $1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 + ax^2 - \frac{1}{6}ax^4 + bx^4 + \cdots = 1$

So $-\frac{1}{6} + a = 0 \therefore a = \frac{1}{6}$

So $\frac{1}{120} - \frac{1}{6} \times \frac{1}{6} + b = 0 \therefore b = \frac{7}{360}$

$\therefore a = \frac{1}{6}$ and $b = \frac{7}{360}$

22) a)

$\mathrm{f}'(x) = 3x^2 \cos(x^3)$

22) b)

Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ then $\cos(x^3) = 1 - \frac{(x^3)^2}{2!} + \frac{(x^3)^4}{4!} - \frac{(x^3)^6}{6!} \ldots$

$3x^2\cos(x^3) \approx 3x^2\left(1 - \frac{1}{2}x^6 + \frac{1}{24}x^{12} - \frac{1}{720}x^{18}\right)$

$= 3x^2 - \frac{3}{2}x^8 + \frac{1}{8}x^{14} - \frac{1}{240}x^{20}$

22) c)

$\int \left(3x^2 - \frac{3}{2}x^8 + \frac{1}{8}x^{14} - \frac{1}{240}x^{20}\right) \mathrm{d}x = x^3 - \frac{1}{6}x^9 + \frac{1}{120}x^{15} - \frac{1}{5040}x^{21} + c$

22) d)

Since $\mathrm{f}(0) = \sin(0) = 0$ and so $c = 0$

$\therefore \sin(x^3) \approx x^3 - \frac{1}{6}x^9 + \frac{1}{120}x^{15} - \frac{1}{5040}x^{21}$

23) a)

$\mathrm{f}(x) = (1-x)^2\mathrm{e}^x$

$\mathrm{f}'(x) = (1-x)^2\mathrm{e}^x - 2(1-x)\mathrm{e}^x = (1 - 2x + x^2 - 2 + 2x)\mathrm{e}^x = (x^2 - 1)\mathrm{e}^x$

$\mathrm{f}''(x) = (x^2 - 1)\mathrm{e}^x + 2x\mathrm{e}^x = (x^2 + 2x - 1)\mathrm{e}^x$ as req.

23) b)

So $\mathrm{f}(0) = 1$, $\mathrm{f}'(0) = -1$, $\mathrm{f}''(0) = -1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

$\therefore (1-x)^2\mathrm{e}^x \approx 1 - x - \frac{1}{2}x^2$

24) a)

$\mathrm{f}(x) = (1-x)\sin x = \sin x - x\sin x$

$\mathrm{f}'(x) = (1-x)\cos x - \sin x$

$\mathrm{f}''(x) = -(1-x)\sin x - \cos x - \cos x = (x-1)\sin x - 2\cos x$ as req.

24) b)

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = -2$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$$

$\therefore (1 - x)\sin x \approx x - x^2$

25) a)

$f(x) = e^{2x}\sin x$ so $f(0) = 0$

$f'(x) = e^{2x}\cos x + 2e^{2x}\sin x$ so $f'(0) = 1$

$\therefore f(0) = 0$ and $f'(0) = 1$

25) b)

$f''(x) = -e^{2x}\sin x + 2e^{2x}\cos x + 2e^{2x}\cos x + 4e^{2x}\sin x$

$= 4e^{2x}\cos x + 3e^{2x}\sin x$

$4f'(x) - 5f(x) = 4(e^{2x}\cos x + 2e^{2x}\sin x) - 5(e^{2x}\sin x)$

$= 4e^{2x}\cos x + 8e^{2x}\sin x - 5e^{2x}\sin x$

$= 4e^{2x}\cos x + 3e^{2x}\sin x$

$\therefore f''(x) = 4f'(x) - 5f(x)$ as req.

25) c)

$f''(0) = 4f'(0) - 5f(0) = 4 - 5 \times 0 = 4$

$f^{(3)}(0) = 4f''(0) - 5f'(0) = 4 \times 4 - 5 \times 1 = 11$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$$

So $e^{2x}\sin x = 0 + x + \frac{4}{2!}x^2 + \frac{11}{3!}x^3 + \cdots$

$\therefore e^{2x}\sin x \approx x + 2x^2 + \frac{11}{6}x^3$

26) a)

$f(x) = e^{-x}\cos x$ so $f(0) = 1$

$f'(x) = -e^{-x}\sin x - e^{-x}\cos x$ so $f'(0) = -1$

$\therefore f(0) = 1$ and $f'(0) = -1$

26) b)

$\mathrm{f}''(x) = -\mathrm{e}^{-x}\cos x + \mathrm{e}^{-x}\sin x + \mathrm{e}^{-x}\sin x + \mathrm{e}^{-x}\cos x$

$= 2\mathrm{e}^{-x}\sin x$

$-2\mathrm{f}'(x) - 2\mathrm{f}(x) = -2(-\mathrm{e}^{-x}\sin x - \mathrm{e}^{-x}\cos x) - 2(\mathrm{e}^{-x}\cos x)$

$= 2\mathrm{e}^{-x}\sin x + 2\mathrm{e}^{-x}\cos x - 2\mathrm{e}^{-x}\cos x$

$= 2\mathrm{e}^{-x}\sin x$

$\therefore \mathrm{f}''(x) = -2\mathrm{f}'(x) - 2\mathrm{f}(x)$ as req.

26) c)

$\mathrm{f}''(0) = -2\mathrm{f}'(0) - 2\mathrm{f}(0) = 2 - 2 = 0$

$\mathrm{f}^{(3)}(0) = -2\mathrm{f}''(0) - 2\mathrm{f}'(0) = 0 + 2 = 2$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\mathrm{e}^{-x}\cos x = 1 - x + 0 + \frac{2}{3!}x^3 + \cdots$

$\therefore \mathrm{e}^{-x}\cos x \approx 1 - x + \frac{1}{3}x^3$

27) a)

$\mathrm{f}(x) = \sin x \cosh x$

$\mathrm{f}'(x) = \sin x \sinh x + \cos x \cosh x$

$\mathrm{f}''(x) = \sin x \cosh x + \cos x \sinh x + \cos x \sinh x - \sin x \cosh x$

$= 2\cos x \sinh x$

$\mathrm{f}^{(3)}(x) = 2\cos x \cosh x - 2\sin x \sinh x$

$\mathrm{f}^{(4)}(x) = 2\cos x \sinh x - 2\sin x \cosh x - 2\sin x \cosh x - 2\cos x \sinh x$

$= -4\sin x \cosh x$

$\therefore \mathrm{f}^{(4)}(x) = -4\mathrm{f}(x)$ so $k = -4$

27) b)

$\mathrm{f}(0) = 0, \mathrm{f}'(0) = 1, \mathrm{f}''(0) = 0, \mathrm{f}^{(3)}(0) = 2, \mathrm{f}^{(4)}(0) = 0$

Since $\mathrm{f}^{(5)}(x) = -4\mathrm{f}'(x)$ then $\mathrm{f}^{(5)}(0) = -4 \times 1 = -4$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\sin x \cosh x = 0 + x + 0 + \frac{2}{3!}x^3 + 0 + \frac{-4}{5!}x^5 + \cdots$

$\therefore \sin x \cosh x \approx x + \frac{1}{3}x^3 - \frac{1}{30}x^5$

28) a)

$f(x) = e^{e^x+1}$ so $f(0) = e^2$

$f'(x) = e^x \times e^{e^x+1} = e^{e^x+1+x}$ so $f'(0) = e^2$

$f''(x) = (e^x + 1)\, e^{e^x+1+x}$ so $f''(0) = 2e^2$

$f^{(3)}(x) = (e^x + 1)^2\, e^{e^x+1+x} + e^{e^x+1+2x}$ so $f^{(3)}(0) = 5\, e^2$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$$

So $e^{e^x+1} = e^2 + e^2x + \frac{2e^2}{2!}x^2 + \frac{5\,e^2}{3!}x^3 + \cdots$

$\therefore e^{e^x+1} \approx e^2 + e^2x + e^2x^2 + \frac{5}{6}\, e^2x^3$

28) b)

$e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots$

So that $e^x + 1 \approx 2 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots$

$\Rightarrow e^{e^x+1} \approx e^{2+x+\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{24}x^4+\cdots} \approx e^2 e^{x+\frac{1}{2}x^2+\frac{1}{6}x^3}$

Consider:

$$e^{x+\frac{1}{2}x^2+\frac{1}{6}x^3} = 1 + \left(x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right) + \frac{1}{2}\left(x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right)^2 + \frac{1}{6}\left(x + \frac{1}{2}x^2 + \frac{1}{6}x^3\right)^3 + \cdots$$

$$= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^3 + \ldots$$

$$\approx 1 + x + x^2 + \frac{5}{6}x^3$$

So that

$$e^2 e^{x+\frac{1}{2}x^2+\frac{1}{6}x^3} = e^2 + e^2x + e^2x^2 + \frac{5e^2}{6}x^3$$

29) a)

$f(x) = \frac{1}{1+e^{-x}}$

$f'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$

$$f''(x) = \frac{-2(1+e^{-x})e^{-x}+(1+e^{-x})^2e^{-x}}{(1+e^{-x})^4} = \frac{-2e^{-x}+(1+e^{-x})e^{-x}}{(1+e^{-x})^3} = \frac{-2e^{-x}+e^{-x}+e^{-2x}}{(1+e^{-x})^3} = \frac{-e^{-x}+e^{-2x}}{(1+e^{-x})^3}$$

$$\frac{-e^{-x}+e^{-2x}}{(1+e^{-x})^3} \times \frac{e^{3x}}{e^{3x}} = \frac{-e^{2x}+e^x}{(e^x+1)^3} = \frac{e^x(1-e^x)}{(e^x+1)^3}$$ as req.

29) b)

$\mathrm{f}(x) = \frac{1}{2}$

$\mathrm{f}'(x) = \frac{1}{4}$

$\mathrm{f}''(x) = 0$

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^x(1-\mathrm{e}^x)\right) = -\mathrm{e}^{2x} + \mathrm{e}^x(1-\mathrm{e}^x) = \mathrm{e}^x - 2\mathrm{e}^{2x}$

$\mathrm{f}^{(3)}(x) = \frac{(\mathrm{e}^x - 2\mathrm{e}^{2x})(\mathrm{e}^x+1)^3 - 3\mathrm{e}^{2x}(1-\mathrm{e}^x)(\mathrm{e}^x+1)^2}{(\mathrm{e}^x+1)^6}$ so $\mathrm{f}^{(3)}(0) = -\frac{1}{8}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\frac{1}{1+\mathrm{e}^{-x}} = \frac{1}{2} + \frac{1}{4}x + 0 + \frac{-\frac{1}{8}}{3!}x^3 + \cdots$

$\therefore \frac{1}{1+\mathrm{e}^{-x}} \approx \frac{1}{2} + \frac{1}{4}x - \frac{1}{48}x^3$

30) a)

$\mathrm{f}(x) = \tan x$

$\mathrm{f}'(x) = \sec^2 x = \frac{1}{\cos^2 x}$

$\mathrm{f}''(x) = \frac{2\sin x \cos x}{\cos^4 x} = 2\tan x \sec^2 x$

$\mathrm{f}^{(3)}(x) = 2\tan x\,(2\tan x \sec^2 x) + (2\sec^2 x)\sec^2 x$

$= 4\tan^2 x \sec^2 x + 2\sec^4 x$

$= 2\sec^2 x\,(\sec^2 x + 2\tan^2 x)$ as req.

30) b)

$\mathrm{f}(0) = 0,\ \mathrm{f}'(0) = 1,\ \mathrm{f}''(0) = 0,\ \mathrm{f}^{(3)}(0) = 2$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\tan x = 0 + x + 0 + \frac{2}{3!}x^3 + \cdots$

$\therefore \tan x \approx x + \frac{1}{3}x^3$

30) c)

$\int_h^{3h} \frac{\tan x}{2x}\,\mathrm{d}x \approx \int_h^{3h} \frac{1}{2x}\left(x + \frac{1}{3}x^3\right)\,\mathrm{d}x = \int_h^{3h} \left(\frac{1}{2} + \frac{1}{6}x^2\right)\,\mathrm{d}x$

$= \left[\frac{1}{2}x + \frac{1}{18}x^3\right]_h^{3h} = \left(\frac{1}{2}(3h) + \frac{1}{18}(3h)^3\right) - \left(\frac{1}{2}(h) + \frac{1}{18}(h)^3\right)$

$= \frac{3}{2}h^3 + \frac{3}{2}h - \frac{1}{18}h^3 - \frac{1}{2}h = h + \frac{13}{9}h^3$ as req.

31) a)

$\mathrm{f}(x) = \ln(\cos x)$

$\mathrm{f}'(x) = -\frac{\sin x}{\cos x} = -\tan x$ as req.

31) b)

$\frac{\mathrm{d}}{\mathrm{d}x}(\sec^2 x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\cos^2 x}\right) = \frac{0+2\cos x \sin x}{\cos^4 x} = 2 \times \frac{\sin x}{\cos x} \times \frac{1}{\cos^2 x} = 2\tan x \sec^2 x$ as req.

31) c)

$\mathrm{f}(x) = \ln(\cos x)$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = -\tan x$ so $\mathrm{f}'(0) = 0$

$\mathrm{f}''(x) = -\sec^2 x = -\frac{1}{\cos^2 x}$ so $\mathrm{f}''(0) = -1$

$\mathrm{f}^{(3)}(x) = -2\tan x \sec^2 x$ so $\mathrm{f}^{(3)}(0) = 0$

$\mathrm{f}^{(4)}(x) = -4\tan^2 x \sec^2 x - 2\sec^4 x$ so $\mathrm{f}^{(4)}(0) = -2$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\ln(\cos x) = 0 + 0 - \frac{1}{2}x^2 + 0 - \frac{2}{24}x^4 + \cdots$

$\therefore \ln(\cos x) \approx -\frac{1}{2}x^2 - \frac{1}{12}x^4$

31) d)

$\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ and so use $x = \frac{\sqrt{3}}{2}$

So $\ln\left(\cos\frac{\pi}{6}\right) \approx -\frac{1}{2}\left(\frac{\pi}{6}\right)^2 - \frac{1}{12}\left(\frac{\pi}{6}\right)^4$

$\therefore \ln\frac{\sqrt{3}}{2} \approx -\frac{\pi^2}{72} - \frac{\pi^4}{15552}$

31) e)

$\frac{\left(-\frac{\pi^2}{72} - \frac{\pi^4}{15552} - \ln\frac{\sqrt{3}}{2}\right)}{\ln\frac{\sqrt{3}}{2}} \times 100 = -0.347\%$

32) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\ln(\sin x)) = \frac{1}{\sin x} \times \cos x = \cot x$

32) b)

$\sin x \approx x - \frac{1}{6}x^3 + \frac{1}{120}x^6$ and $\ln(1+x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

$\ln(\sin x) \approx \ln\left(x - \frac{1}{6}x^3 + \frac{1}{120}x^6\right)$

$= \ln\left(x\left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^5\right)\right) = \ln x + \ln\left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4\right)$

$\approx \ln x + \left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right) - \frac{1}{2}\left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right)^2 + \frac{1}{3}\left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right)^3$

$= \ln x + \left(-\frac{1}{6}x^2 + \frac{1}{120}x^4\right) - \frac{1}{2}\left(\frac{1}{36}x^4 - \cdots\right) + O(x^6)$

$= \ln x - \frac{1}{6}x^2 - \frac{1}{180}x^4 + O(x^6)$

$\therefore \ln(\sin x) = \ln x - \frac{1}{6}x^2 - \frac{1}{180}x^4$

So differentiating gives:

$\cot x \approx \frac{1}{x} - \frac{1}{3}x - \frac{1}{45}x^3$

33) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\tan\left(\frac{x}{2}\right)\right)\right) = \frac{1}{\tan\left(\frac{x}{2}\right)} \times \sec^2\left(\frac{x}{2}\right) \times \frac{1}{2} = \frac{\sec^2\left(\frac{x}{2}\right)}{2\tan\left(\frac{x}{2}\right)} = \frac{\cos\left(\frac{x}{2}\right)}{2\sin\left(\frac{x}{2}\right)\cos^2\left(\frac{x}{2}\right)} = \frac{1}{2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}$

Since $2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right) = \sin x$

So $\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\tan\left(\frac{x}{2}\right)\right)\right) = \frac{1}{\sin x} = \operatorname{cosec} x$ as req.

33) b)

$\tan\left(\frac{x}{2}\right) = \left(\frac{x}{2}\right) + \frac{\left(\frac{x}{2}\right)^3}{3} + \frac{2\left(\frac{x}{2}\right)^5}{15} = \frac{x}{2} + \frac{x^3}{24} + \frac{x^5}{240}$

$\ln\left(\tan\left(\frac{x}{2}\right)\right) \approx \ln\left(\frac{x}{2} + \frac{x^3}{24} + \frac{x^5}{240}\right)$

Using the expansion for $\ln(1+x)$:

$\ln\left(\frac{x}{2}\left(1 + \frac{x^2}{12} + \frac{x^4}{120}\right)\right) = \ln\left(\frac{x}{2}\right) + \ln\left(1 + \frac{x^2}{12} + \frac{x^4}{120}\right)$

$= \ln\left(\frac{x}{2}\right) + \left(\frac{x^2}{12} + \frac{x^4}{120}\right) - \frac{1}{2}\left(\frac{x^2}{12} + \frac{x^4}{120}\right)^2 + \cdots$

$\approx \ln\left(\frac{x}{2}\right) + \frac{x^2}{12} + \frac{7x^4}{1440}$

Since $\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\tan\left(\frac{x}{2}\right)\right)\right) = \operatorname{cosec} x$ then

$\operatorname{cosec} x \approx \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{x}{2}\right) + \frac{x^2}{12} + \frac{7x^4}{1440}\right) = \frac{1}{x} + \frac{x}{6} + \frac{7x^3}{360}$

34) a)

$y = \arcsin x \Rightarrow \sin y = x$

$\cos y \frac{\mathrm{d}y}{\mathrm{d}x} = 1 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cos y}$

But since $\sin y = \frac{x}{1}$ then $\cos y = \sqrt{1 - x^2}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}$

$\therefore \mathrm{f}'(x) = \frac{1}{\sqrt{1-x^2}}$ as req.

34) b)

$\mathrm{f}(x) = \arcsin x$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = \frac{1}{\sqrt{1-x^2}}$ so $\mathrm{f}'(0) = 1$

$\mathrm{f}''(x) = x(1 - x^2)^{-\frac{3}{2}} = \frac{x}{(1-x^2)^{\frac{3}{2}}}$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \frac{(1-x^2)^{\frac{3}{2}} + 3x^2(1-x^2)^{\frac{1}{2}}}{(1-x^2)^3}$ so $\mathrm{f}^{(3)}(0) = 1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arcsin x = 0 + x + 0 + \frac{1}{6}x^3 + \cdots$

$\therefore \arcsin x \approx x + \frac{1}{6}x^3$

35) a)

$y = \arccos x \Rightarrow \cos y = x$

$-\sin y \frac{\mathrm{d}y}{\mathrm{d}x} = 1 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sin y}$

But since $\cos y = \frac{x}{1}$ then $\sin y = \sqrt{1 - x^2}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}}$

$\therefore \mathrm{f}'(x) = -\frac{1}{\sqrt{1-x^2}}$ as req.

35) b)

$\mathrm{f}(x) = \arccos x$ so $\mathrm{f}(0) = \frac{\pi}{2}$

$\mathrm{f}'(x) = -\frac{1}{\sqrt{1-x^2}}$ so $\mathrm{f}'(0) = -1$

$\mathrm{f}''(x) = -x(1-x^2)^{-\frac{3}{2}} = -\frac{x}{(1-x^2)^{\frac{3}{2}}}$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \frac{-(1-x^2)^{\frac{3}{2}} + 3x^2(1-x^2)^{\frac{1}{2}}}{(1-x^2)^3}$ so $\mathrm{f}^{(3)}(0) = -1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arccos x = \frac{\pi}{2} - x + 0 - \frac{1}{6}x^3 + \cdots$

$\therefore \arccos x \approx \frac{\pi}{2} - x - \frac{1}{6}x^3$

36) a)

$y = \arctan x \Rightarrow \tan y = x$

$\sec^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sec^2 y} = \cos^2 y$

But since $\tan y = \frac{x}{1}$ then $\cos y = \frac{1}{\sqrt{1+x^2}}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{1}{\sqrt{1+x^2}}\right)^2$

$\therefore \mathrm{f}'(x) = \frac{1}{1+x^2}$ as req.

36) b)

$\mathrm{f}(x) = \arctan x$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = \frac{1}{1+x^2}$ so $\mathrm{f}'(0) = 1$

$\mathrm{f}''(x) = \frac{-2x}{(1+x^2)^2}$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \frac{-2(1+x^2)^2 + 8x^2(1+x^2)}{(1+x^2)^4} = \frac{-2(1+x^2)^2 + 8x^2}{(1+x^2)^3} = \frac{-2+6x^2}{(1+x^2)^3}$ so $\mathrm{f}^{(3)}(0) = -2$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arctan x = 0 + x + 0 - \frac{2}{6}x^3 + \cdots$

$\therefore \arctan x \approx x - \frac{1}{3}x^3$

36) c)

Since $\arctan 1 = \frac{\pi}{4}$ then

$\arctan 1 = 1 - \frac{1}{3} + \cdots$

So $\frac{\pi}{4} = 1 - \frac{1}{3} + \cdots$

37) a)

$\frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}} = 1 + \left(-\frac{1}{2}\right)(x^2) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(x^2)^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(x^2)^3 + \cdots$

$\approx 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6$

37) b)

Since $\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arsinh} x) = \frac{1}{\sqrt{1+x^2}}$

So $\operatorname{arsinh} x = \int \frac{1}{\sqrt{1+x^2}}\,\mathrm{d}x \approx \int \left(1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6\right)\mathrm{d}x$

$\therefore \operatorname{arsinh} x \approx x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7$

38)

Since $\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{artanh} x) = \frac{1}{1-x^2}$

$\frac{1}{1-x^2} = (1+x^2)^{-1} = 1 + (-1)(-x^2) + \frac{(-1)(-2)}{2!}(-x^2)^2 + \frac{(-1)(-2)(-3)}{3!}(-x^2)^3 + \cdots$

$= 1 + x^2 + x^4 + x^6 + \cdots$

So $\operatorname{artanh} x = \int \frac{1}{1+x^2}\,\mathrm{d}x \approx \int (1 + x^2 + x^4 + x^6)\,\mathrm{d}x$

$\therefore \operatorname{artanh} x \approx x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7$ as req.

39) a)

$\mathrm{f}(x) = \arccos x$ so $\mathrm{f}(0) = \frac{\pi}{2}$

$\mathrm{f}'(x) = -\frac{1}{\sqrt{1-x^2}}$ so $\mathrm{f}'(0) = -1$

$\mathrm{f}''(x) = -x(1-x^2)^{-\frac{3}{2}} = -\frac{x}{(1-x^2)^{\frac{3}{2}}}$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \frac{-(1-x^2)^{\frac{3}{2}} + 3x^2(1-x^2)^{\frac{1}{2}}}{(1-x^2)^3}$ so $\mathrm{f}^{(3)}(0) = -1$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arccos x = \frac{\pi}{2} - x + 0 - \frac{1}{6}x^3 + \cdots$

$\therefore \arccos x \approx \frac{\pi}{2} - x - \frac{1}{6}x^3$

39) b)

Since $\ln(1+x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots$

Then $(\arccos x)\ln(1+x) \approx \left(\frac{\pi}{2} - x - \frac{1}{6}x^3\right)\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right)$

$= \frac{\pi}{2}x - \frac{\pi}{4}x^2 + \frac{\pi}{6}x^3 - x^2 + \frac{1}{2}x^3 + \cdots$

$= \frac{\pi}{2}x + (-1 - \frac{\pi}{4})x^2 + \left(\frac{1}{2} + \frac{\pi}{6}\right)x^3 + \cdots$

So that

The expansion of $(\arccos x)\ln(1+x)$ up to and including the x^3 term is:

$\frac{\pi}{2}x + \left(-1 - \frac{\pi}{4}\right)x^2 + \left(\frac{1}{2} + \frac{\pi}{6}\right)x^3 = \frac{\pi}{2}x + \left(-1 - \frac{\pi}{4}\right)x^2 + \frac{1}{6}(3+\pi)x^3$ as req.

40) a)

$\mathrm{f}(x) = \arctan(1+x)$ so $\mathrm{f}(0) = \frac{\pi}{4}$

$\mathrm{f}'(x) = \frac{1}{1+(1+x)^2}$ so $\mathrm{f}'(0) = \frac{1}{2}$

$\mathrm{f}''(x) = \frac{-2(1+x)}{(1+(1+x)^2)^2}$ so $\mathrm{f}''(0) = -\frac{1}{2}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arctan(1+x) = \frac{\pi}{4} + \frac{1}{2}x - \frac{\frac{1}{2}}{2!}x^2 + \cdots$

$\therefore \arctan(1+x) \approx \frac{\pi}{4} + \frac{1}{2}x - \frac{1}{4}x^2$

40) b)

$\int_{-h}^{h} x \arctan(1+x)\ \mathrm{d}x \approx \int_{-h}^{h} x\left(\frac{\pi}{4}+\frac{1}{2}x-\frac{1}{4}x^2\right)\mathrm{d}x = \int_{-h}^{h}\left(\frac{\pi}{4}x+\frac{1}{2}x^2-\frac{1}{4}x^3\right)\mathrm{d}x$

$= \left[\frac{\pi}{8}x^2+\frac{1}{6}x^3-\frac{1}{16}x^4\right]_{-h}^{h}$

$= \left(\frac{\pi}{8}(h)^2+\frac{1}{6}(h)^3-\frac{1}{16}(h)^4\right)-\left(\frac{\pi}{8}(-h)^2+\frac{1}{6}(-h)^3-\frac{1}{16}(-h)^4\right)$

$= \frac{1}{6}h^3+\frac{1}{6}h^3 = \frac{1}{3}h^3$ as req.

41) a)

$\mathrm{f}(x) = \arcsin x$ so $\mathrm{f}(0) = 0$

$\mathrm{f}'(x) = \frac{1}{\sqrt{1-x^2}}$ so $\mathrm{f}'(0) = 1$

$\mathrm{f}''(x) = x(1-x^2)^{-\frac{3}{2}} = \frac{x}{(1-x^2)^{\frac{3}{2}}}$ so $\mathrm{f}''(0) = 0$

$\mathrm{f}^{(3)}(x) = \frac{(1-x^2)^{\frac{3}{2}}+3x^2(1-x^2)^{\frac{1}{2}}}{(1-x^2)^3}$ so $\mathrm{f}^{(3)}(0) = 1$

$\mathrm{f}(x) = \mathrm{f}(0)+\mathrm{f}'(0)x+\frac{\mathrm{f}''(0)}{2!}x^2+\frac{\mathrm{f}^{(3)}(0)}{3!}x^3+\frac{\mathrm{f}^{(4)}(0)}{4!}x^4+\frac{\mathrm{f}^{(5)}(0)}{5!}x^5+\cdots+\frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arcsin x = 0+x+0+\frac{1}{6}x^3+\cdots$

$\therefore \arcsin x \approx x+\frac{1}{6}x^3$

41) b)

$\int_0^{\frac{1}{2}} \arcsin x\,\mathrm{d}x \approx \int_0^{\frac{1}{2}}\left(x+\frac{1}{6}x^3\right)\mathrm{d}x \approx \left[\frac{1}{2}x^2+\frac{1}{24}x^4\right]_0^{\frac{1}{2}} = \left(\frac{1}{8}+\frac{1}{384}\right)-(0+0) = \frac{49}{384}$

$= 0.128$ (3 SF)

41) c)

$\int \arcsin x \, \mathrm{d}x = \int 1 \times \arcsin x \, \mathrm{d}x$

Use Integration by Parts:

$u = \arcsin x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}} \qquad v = x$

$\therefore \int \arcsin x \, \mathrm{d}x = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$ use Integration by Substitution:

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = -u^{\frac{1}{2}} + c = -\sqrt{1-x^2} + c$

So:

$\int \arcsin x \, \mathrm{d}x = x \arcsin x + \sqrt{1-x^2} + c$

So that $\int_0^{\frac{1}{2}} \arcsin x \, \mathrm{d}x = \left[x \arcsin x + \sqrt{1-x^2}\right]_0^{\frac{1}{2}} = \left(\frac{\pi}{12} + \frac{\sqrt{3}}{2}\right) - (0 + 1) = \frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}$

$\therefore \int_0^{\frac{1}{2}} \arcsin x \, \mathrm{d}x = \frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}$

41) d)

$\frac{\left(\frac{49}{384} - \left(\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}\right)\right)}{\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}} \times 100 = -0.2\%$ (1 DP)

42) a)

$\mathrm{f}(x) = \arcsin\left(x + \frac{1}{3}\right)$ so $\mathrm{f}(0) = \arcsin\left(\frac{1}{3}\right)$

$\mathrm{f}'(x) = \frac{1}{\sqrt{1-\left(x+\frac{1}{3}\right)^2}}$ so $\mathrm{f}'(0) = \frac{3\sqrt{2}}{4}$

$\mathrm{f}''(x) = \left(x + \frac{1}{3}\right)\left(1 - \left(x + \frac{1}{3}\right)^2\right)^{-\frac{3}{2}}$ so $\mathrm{f}''(0) = \frac{9\sqrt{2}}{32}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arcsin\left(x + \frac{1}{3}\right) = \arcsin\left(\frac{1}{3}\right) + \frac{3\sqrt{2}}{4}x + \frac{\frac{9\sqrt{2}}{32}}{2}x^2 + \cdots$

$\therefore \arcsin\left(x + \frac{1}{3}\right) \approx \arcsin\left(\frac{1}{3}\right) + \frac{3\sqrt{2}}{4}x + \frac{9\sqrt{2}}{64}x^2$

42) b)

$\int_0^{0.05} \arcsin\left(x + \frac{1}{3}\right) \mathrm{d}x \approx \int_0^{0.05} \left(\arcsin\left(\frac{1}{3}\right) + \frac{3\sqrt{2}}{4}x + \frac{9\sqrt{2}}{64}x^2\right) \mathrm{d}x$

$= \left[\arcsin\left(\frac{1}{3}\right)x + \frac{3\sqrt{2}}{8}x^2 + \frac{3\sqrt{2}}{64}x^3\right]_0^{0.05}$

$= \left(\arcsin\left(\frac{1}{3}\right)(0.05) + \frac{3\sqrt{2}}{8}(0.05)^2 + \frac{3\sqrt{2}}{64}(0.05)^3\right) - (0)$

$= 0.0183$ (3 SF)

43) a)

$\mathrm{f}(x) = \arccos(2x)$ so $\mathrm{f}'(x) = -\frac{2}{\sqrt{1-4x^2}}$

43) b)

Use Binomial Series:

$-2(1-4x^2)^{-\frac{1}{2}} = -2\left(1 + \left(-\frac{1}{2}\right)(-4x^2) + \frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2}(-4x^2)^2 + \cdots\right)$

$= -2 - 4x^2 - 12x^4 + \cdots$

$\int -\frac{2}{\sqrt{1-4x^2}}\mathrm{d}x = \int \arccos(2x)\,\mathrm{d}x$

So $\int -\frac{2}{\sqrt{1-4x^2}}\mathrm{d}x \approx \int (-2 - 4x^2 - 12x^4)\mathrm{d}x = -2x - \frac{4}{3}x^3 - \frac{12}{5}x^5 + c$

Since $\arccos(0) = \frac{\pi}{2}$ so $\frac{\pi}{2} = 0 + c$ gives $c = \frac{\pi}{2}$

$\therefore \arccos(2x) \approx \frac{\pi}{2} - 2x - \frac{4}{3}x^3 - \frac{12}{5}x^5$

44) a)

$f(x) = \arcsin(2x)$ so $f(0) = 0$

$f'(x) = \frac{2}{\sqrt{1-4x^2}} = 2(1-4x^2)^{-\frac{1}{2}}$ so $f'(0) = 2$

$f''(x) = 8x(1-4x^2)^{-\frac{3}{2}}$ so $f''(0) = 0$

$f^{(3)}(x) = 64x^2(1-4x^2)^{-\frac{5}{2}} + 8(1-4x^2)^{-\frac{3}{2}}$ so $f^{(3)}(0) = 8$

$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}}{r!}x^r$

So $\arcsin(2x) = 0 + 2x + 0 + \frac{8}{3!}x^3 + \cdots$

$\therefore \arcsin(2x) \approx 2x + \frac{4}{3}x^3$

44) b)

When $x = \frac{1}{4}$, $\arcsin\left(2 \times \frac{1}{4}\right) = \frac{\pi}{6}$

So $\frac{\pi}{6} \approx 2 \times \frac{1}{4} + \frac{4}{3} \times \left(\frac{1}{4}\right)^3$

$\frac{\pi}{6} \approx \frac{1}{2} + \frac{1}{48}$

$\therefore \pi \approx 6\left(\frac{1}{2} + \frac{1}{48}\right)$ as req.

44) c)

$\frac{6\left(\frac{1}{2}+\frac{1}{48}\right)-\pi}{\pi} \times 100 = -0.528\%$

45) a)

$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}$ so $\frac{d}{dx}(\operatorname{artanh} u) = \frac{\frac{du}{dx}}{1-u^2}$

$u = \frac{1-x}{3+x}$ so $\frac{du}{dx} = \frac{-(3+x)-(1-x)}{(3+x)^2} = -\frac{4}{(3+x)^2}$

$1 - u^2 = 1 - \left(\frac{1-x}{3+x}\right)^2 = 1 - \frac{(1-x)^2}{(3+x)^2} = \frac{(3+x)^2}{(3+x)^2} - \frac{(1-x)^2}{(3+x)^2} = \frac{(3+x)^2-(1-x)^2}{(3+x)^2} = \frac{9+6x+x^2-1+2x-x^2}{(3+x)^2}$

$= \frac{8+8x}{(3+x)^2}$

So that $f'(x) = \frac{-\frac{4}{(3+x)^2}}{\frac{8+8x}{(3+x)^2}} = -\frac{4}{8+8x} = -\frac{1}{2x+2}$ as req.

45) b)

$\mathrm{f}''(x) = \frac{2}{(2x+2)^2} = \frac{2}{4(x+1)^2} = \frac{1}{2(x+1)^2}$

45) c)

$\mathrm{f}(x) = \operatorname{artanh}\left(\frac{1-x}{3+x}\right)$ so $\mathrm{f}(0) = \operatorname{artanh}\left(\frac{1}{3}\right) = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$

$\mathrm{f}'(x) = -\frac{1}{2x+2}$ so $\mathrm{f}'(0) = -\frac{1}{2}$

$\mathrm{f}''(x) = \frac{1}{2(x+1)^2}$ so $\mathrm{f}''(0) = \frac{1}{2}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\operatorname{artanh}\left(\frac{1-x}{3+x}\right) = \frac{1}{2}\ln 2 - \frac{1}{2}x + \frac{\frac{1}{2}}{2!}x^2 + \cdots$

$\therefore \operatorname{artanh}\left(\frac{1-x}{3+x}\right) \approx \frac{1}{2}\ln 2 - \frac{1}{2}x + \frac{1}{4}x^2$ as req.

46) a)

$\mathrm{f}(x) = \arctan(\cosh x)$

$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2}$ so $\frac{\mathrm{d}}{\mathrm{d}x}(\arctan u) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}}{1+u^2}$

$u = \cosh x$ so $\frac{\mathrm{d}u}{\mathrm{d}x} = \sinh x$

So that $\mathrm{f}'(x) = \frac{\sinh x}{1+\cosh^2 x}$

$\mathrm{f}''(x) = \frac{\cosh x(1+\cosh^2 x) - 2\sinh^2 x \cosh x}{(1+\cosh^2 x)^2} = \frac{\cosh x(1+\cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}$

46) b)

$\mathrm{f}(x) = \arctan(\cosh x)$ so $\mathrm{f}(0) = \frac{\pi}{4}$

$\mathrm{f}'(x) = \frac{\sinh x}{1+\cosh^2 x}$ so $\mathrm{f}'(0) = 0$

$\mathrm{f}''(x) = \frac{\cosh x(1+\cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}$ so $\mathrm{f}''(0) = \frac{1}{2}$

$\mathrm{f}(x) = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2!}x^2 + \frac{\mathrm{f}^{(3)}(0)}{3!}x^3 + \frac{\mathrm{f}^{(4)}(0)}{4!}x^4 + \frac{\mathrm{f}^{(5)}(0)}{5!}x^5 + \cdots + \frac{\mathrm{f}^{(r)}}{r!}x^r$

So $\arctan(\cosh x) = \frac{\pi}{4} + 0 + \frac{\frac{1}{2}}{2!}x^2 + \cdots$

$\therefore \arctan(\cosh x) \approx \frac{\pi}{4} + \frac{1}{4}x^2$